

and smart watches. That makes it difficult to form a collaboration across each of them.

3. It gets hard for application developers to develop an application for different platforms that use different operating systems. They have to learn about such platforms along with different sets of artifacts and operating systems.

What does HarmonyOS do?

HarmonyOS creates a resource pool (as shown in Fig. 1) across all the devices and using the distributed virtual bus it tries to connect them all. It helps application developers to combine the devices into a single super device. It allows the developed applications to use the resources from all the devices that are available, which is very useful for the end customer.

It is denoted as 1 + 8 + N strategy (refer Fig. 2) by Huawei where '1' focuses on devices such as smartphones, '8' covers the majority of the devices such as eyewear, tablet, PC (personal computer), watch, speakers, vehicle terminal, earphones, TVs (television), and 'N' represents all the available IoT devices. It is backed and developed by Huawei. HarmonyOS currently supports 1 + 8 devices, which are backed and developed by Huawei itself.

HarmonyOS user scenarios

Based on the Harmony architecture you do not have to install any other application on your mobile in order to operate other devices. That is automatically done by the atomic service. It will initially complete the Wi-Fi provisioning and then you can start using it.

Fitness and health. You can use your mobile phone for cross-device resource convergence. There are devices like fitness bands, phones, weight trackers, and treadmills that can be collaborated into one super device with a simple integration of

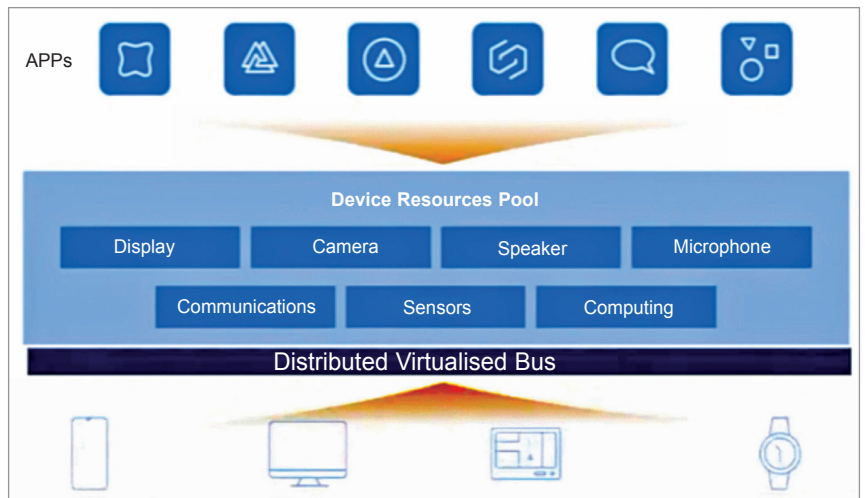


Fig. 1: Application pool



Fig. 2: List of devices under Huawei's 1+8+N strategy

virtualisation that uses HarmonyOS as its backbone.

Smart home. Home appliances can easily be operated just with a touch of a button. You can preheat your oven and choose the right configurations to prepare a particular dish. You can receive notifications on your device that tell you when you should be changing the filter of your air-purifier. It even allows you to seamlessly receive customer care services and lets you get an appointment for the same.

It is important to note that these applications do not come pre-installed. Depending on the services you need and the type of device you have, the corresponding UI (user interface) is provided. The experience remains the same across all the devices.

Entertainment. For entertainment, users can stream the desired content onto bigger screens, they

can use smart speakers and integrate a microphone, which allows them to have a collaborative experience. Again, all of this is controlled by a single device by the end user. They get to control all the components as per their need.

Smart office. Nowadays, most people are working from their homes. It is often difficult for people to move from one corner to another with their laptops when they have to. HarmonyOS gives them the flexibility to integrate all of their devices, such as TVs, phones, laptops, and tablets. This integration allows them to work from any device, and they can access all the files from any device and update them as well from anywhere they want. It improves convenience and increases productivity.

Easy travel. Sometimes while navigating your way to a certain destination you do not reach your